STRUCTURE, RESPONSE AND THE RETURN OF THE OSCILLATOR

How equations gain sophistication

The unknown grows from a value to a trajectory, a field and a spectrum of modes.

1 / RELATE QUANTITIES AT A SPECIFIED STATE

Algebra: find an unknown value

A relation lets us calculate one quantity from others: for example, $V = RI$ gives $I = V/R$.

Model	Relation	Role
Spring	$F_\mathrm{s} = -kx$	Response law
Resistor	$V = RI$	Response law
Ideal gas	$pV = nR_\mathrm{g}T$	Equation of state

The first two are **constitutive laws**: they specify a response within a model's range of validity. Here k, R describe the particular spring or resistor; R_g is the universal gas constant.

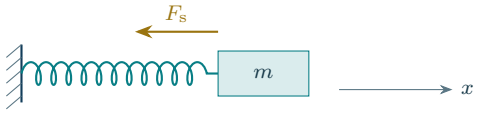
2 / SEPARATE THE FRAMEWORK FROM THE RESPONSE

A structural law supplies the balance

For one coordinate and constant mass, Newton's law is

$$\sum_i F_i \quad = \quad m\ddot{x}.$$

The sum collects the forces; response laws supply them. Dots are time derivatives: $\dot{x} = v$, $\ddot{x} = a$.



3 / COMBINE STRUCTURE WITH CONSTITUTION

An ODE: find an entire trajectory

For a free, frictionless spring, measure x from equilibrium. Insert $F_\mathrm{s} = -kx$ into Newton's law:

$$m\ddot{x} + kx \quad = \quad 0, \qquad \omega_0^2 \quad = \quad \frac{k}{m}.$$

This is an **ordinary differential equation**: the unknown is a function $x(t)$, rather than a single value. For $m, k > 0$, its solutions are

$$x(t) = x_0 \cos(\omega_0 t) + \frac{v_0}{\omega_0} \sin(\omega_0 t).$$

The two initial conditions $x(0) = x_0$, $\dot{x}(0) = v_0$ choose one motion from the family.

4 / GIVE EVERY POSITION ITS OWN VARIABLE

A PDE: find a whole field

A uniform, taut string has displacement $u(z,t)$: one value at each position and time. For small transverse waves,

$$\frac{\partial^2 u}{\partial t^2} \quad = \quad c^2 \frac{\partial^2 u}{\partial z^2}, \qquad c^2 \quad = \quad \frac{\mathcal{T}}{\mu}.$$

Here $\mathcal{T}$ is tension, μ is mass per unit length. A **partial differential equation** relates changes in time and space. For fixed ends,

$$u(0,t) = u(L,t) = 0.$$

Supply two initial profiles: the shape $u(z,0)$ and the velocity $u_t(z,0)$, as well as these boundary conditions.

5 / THE SAME ORGANISING PATTERN IN SPACE

Field equations also need a response law

In electrostatics, use potential ϕ, electric field $\mathbf{E}$, and electric displacement $\mathbf{D}$. For a linear, isotropic dielectric:

$$\mathbf{E} = -\nabla\phi \quad \text{potential relation,}$$
$$\nabla\cdot\mathbf{D} = \rho_\mathrm{f} \quad \text{Gauss's law,}$$
$$\mathbf{D} = \varepsilon\mathbf{E} \quad \text{constitutive law.}$$

The free-charge density ρ_f supplies the source; the permittivity ε specifies the electric response. Combining the three equations gives

$$\nabla \cdot \big(\varepsilon\nabla\phi\big) \quad = \quad -\rho_\mathrm{f}.$$

For constant ε, this is **Poisson's equation**:

$$\nabla^2\phi = -\frac{\rho_\mathrm{f}}{\varepsilon}.$$

The gradient $\nabla\phi$ measures spatial change; the Laplacian ∇^2 sums second spatial derivatives.

Specify ρ_f, ε and the boundary conditions.

6 / ASK WHICH SHAPES AND FREQUENCIES ARE ALLOWED

An eigenvalue problem: find the modes

For the fixed string, the spatial shapes solve

$$-\psi_n'' = \lambda_n\psi_n, \qquad \psi_n(0) = \psi_n(L) = 0.$$

Non-zero solutions occur at discrete eigenvalues:

$$\lambda_n = \left(\frac{n\pi}{L}\right)^2, \qquad \psi_n(z) = \sin\frac{n\pi z}{L}, \quad n \geq 1.$$



Write the field as $u(z,t) = \sum_{n\geq 1} q_n(t)\psi_n(z)$. Each mode amplitude then obeys

$$\ddot{q}_n + \omega_n^2 q_n \quad = \quad 0, \qquad \omega_n \quad = \quad \frac{n\pi c}{L}.$$

The field has become a collection of harmonic oscillators. The boundary conditions select the spectrum; the initial profiles fix the mode amplitudes and phases.

7 / QUANTISE THE MODES

For a free bosonic field, each independent mode has discrete oscillator energies. N counts the quanta in mode n.

$$E_{n,N} = \hbar\omega_n\left(N + \tfrac{1}{2}\right)$$
$$N = 0, 1, 2, \ldots$$

Reading an equation: identify the unknown, the structural law, the response model, and the data needed to select a solution.

Explore: *On the Structure of Equations*. **Sources:** Feynman II.12; MIT 8.03, Lecture 9; Tong, *Quantum Field Theory*, §2.

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